Data and Artificial Intelligence

Cyber Shujaa Program

Week 3 Assignment
Titanic Exploratory Data Analysis

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1. Introduction

This assignment focused on performing Exploratory Data Analysis (EDA) using the Titanic dataset from Kaggle.
The purpose was to gain practical experience in profiling data, handling missing values, analyzing distributions, detecting outliers, and exploring feature relationships that influence survival on the Titanic.

2. Objectives

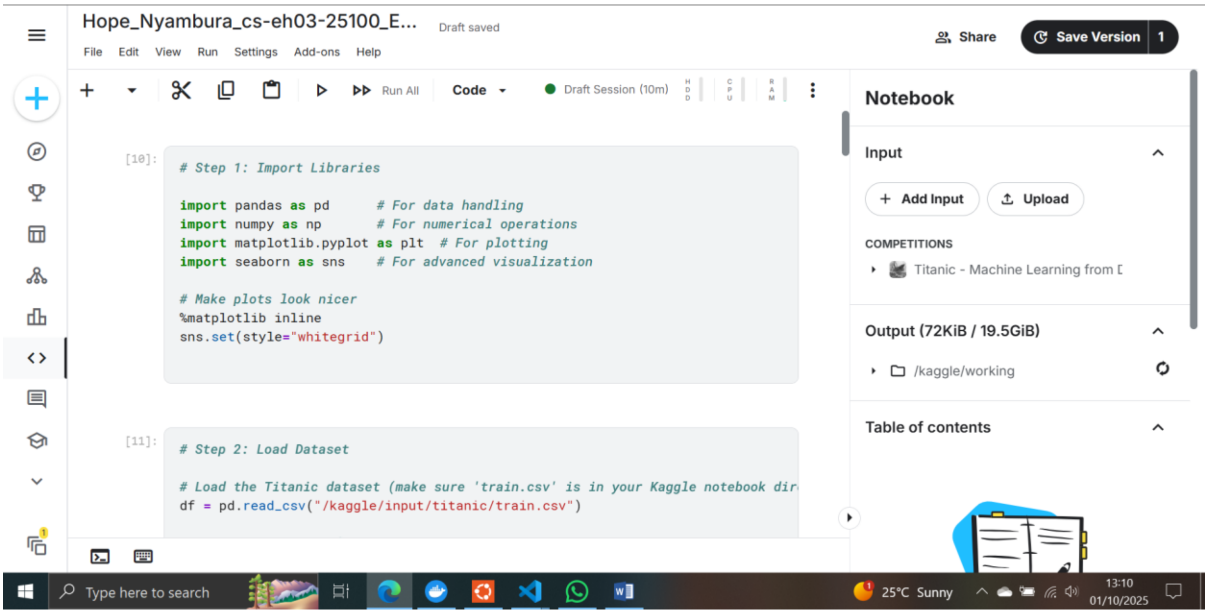
The main objectives of this assignment were:

- To explore the Titanic dataset and understand its structure.
- To assess and handle missing values, duplicates, and outliers.
- To conduct univariate, bivariate, and multivariate analysis.
- To analyze the survival variable and identify key factors influencing survival.
- To publish the final notebook on Kaggle.

3. Tasks Completed

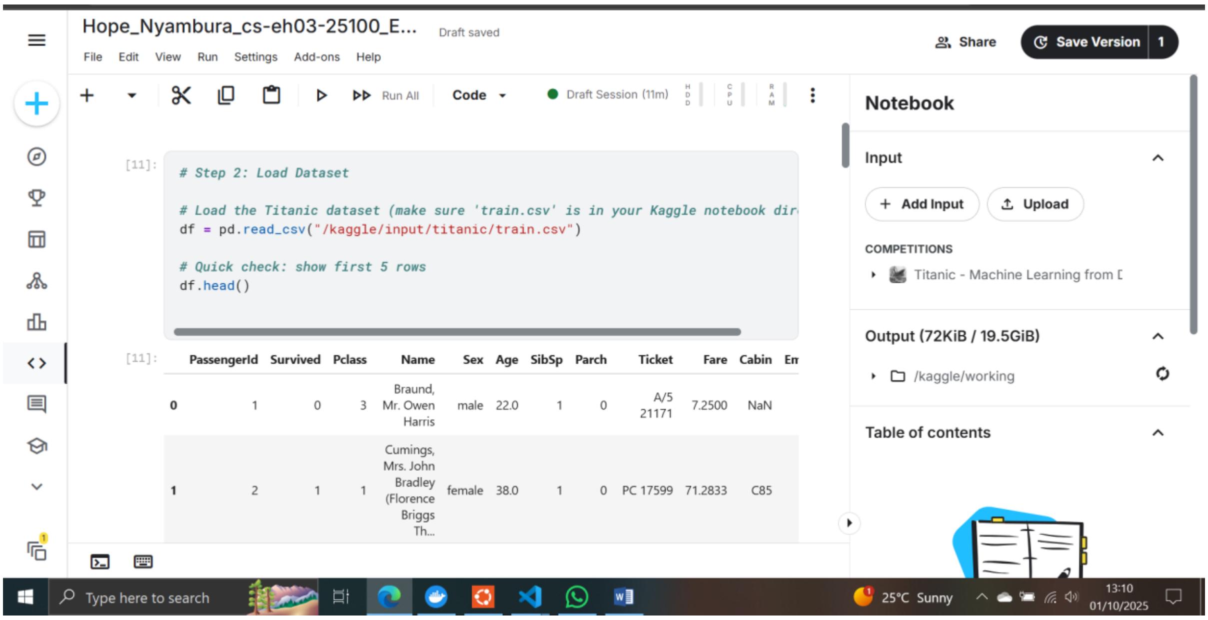
Step 1: Import Libraries

- Imported Pandas, NumPy, Matplotlib, and Seaborn for analysis and visualization.



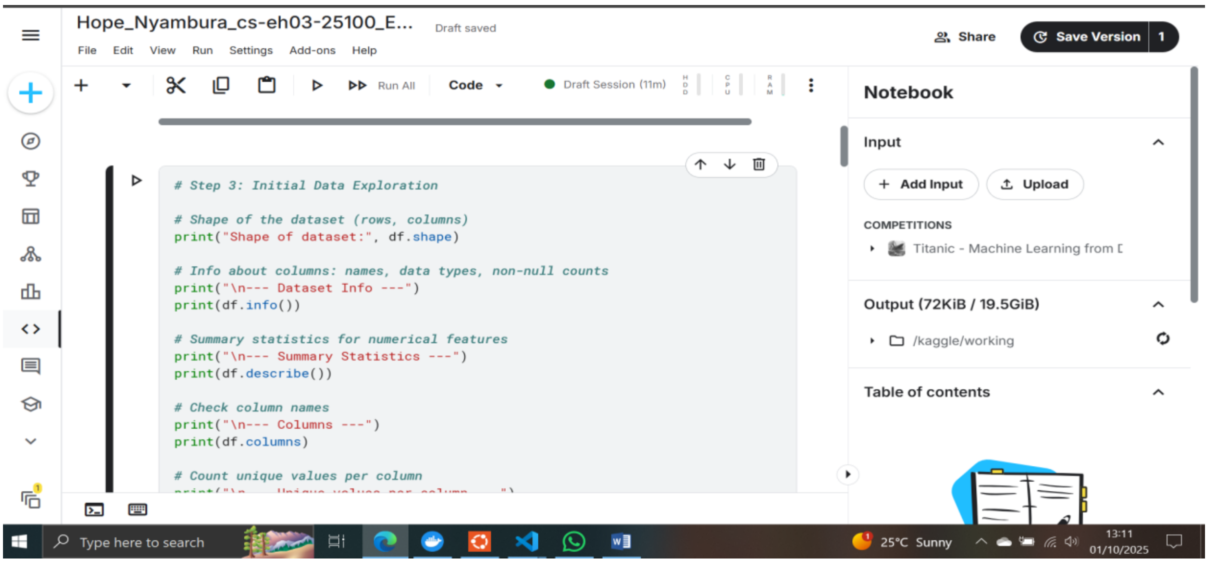
Step 2: Load Dataset

- Loaded the Titanic dataset into a Pandas DataFrame.
- Displayed the first few rows using df.head().



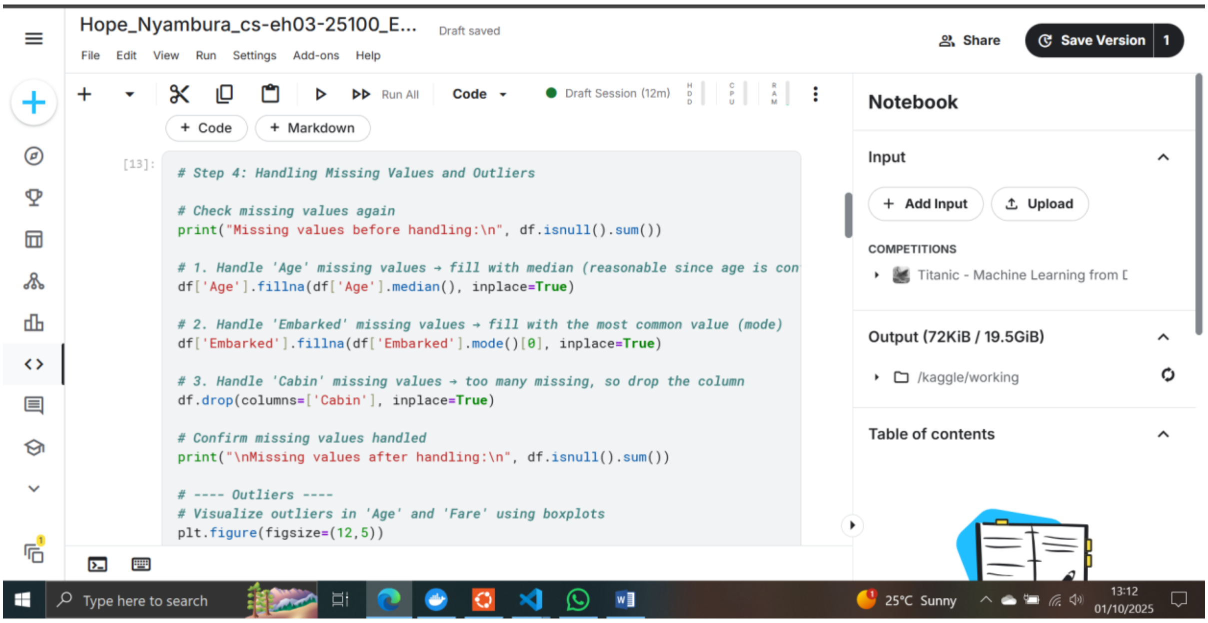
Step 3: Initial Data Exploration

- Used:
    - df.shape → to get rows and columns.
    - df.info() → to check data types and missing values.
    - df.describe() → to summarize numerical features.
    - df.duplicated().sum() → checked duplicate rows.
    - df.nunique() → counted unique values per column.
- Documented feature list (PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked).



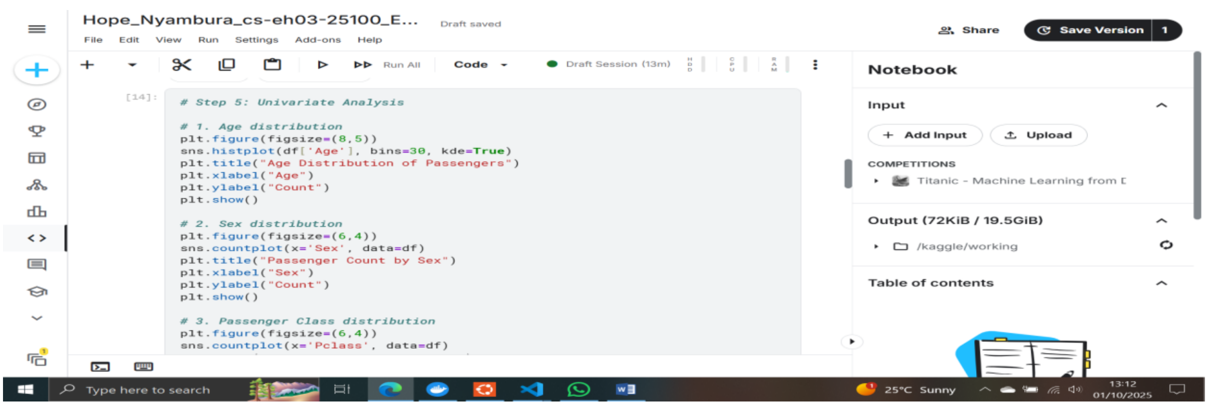
Step 4: Handling Missing Values and Outliers

- Missing values:
  - Age → imputed with median.
  - Cabin → dropped due to high missingness.
  - Embarked → filled with most common port.
- Outliers:
  - Detected using boxplots for Age and Fare.
  - Decided to keep extreme fares (important for wealth insights).



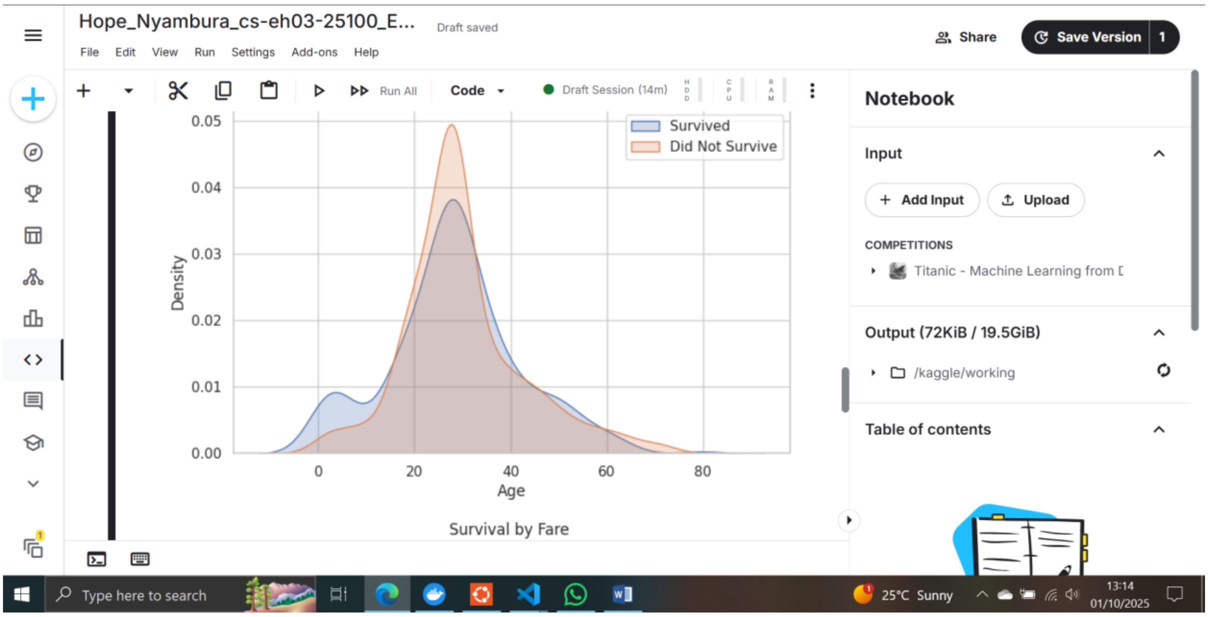
Step 5: Univariate Analysis

- Age distribution → histogram.
- Sex counts → bar plot.
- Pclass distribution → count plot.
- Fare distribution → histogram (skewed).
- Embarked distribution → count plot.



Step 6: Bivariate Analysis

- Survival vs Sex → count plot.
- Survival vs Pclass → bar plot.
- Survival vs Age → KDE plot.
- Survival vs Embarked → bar plot.
- Survival vs Fare → violin plot.

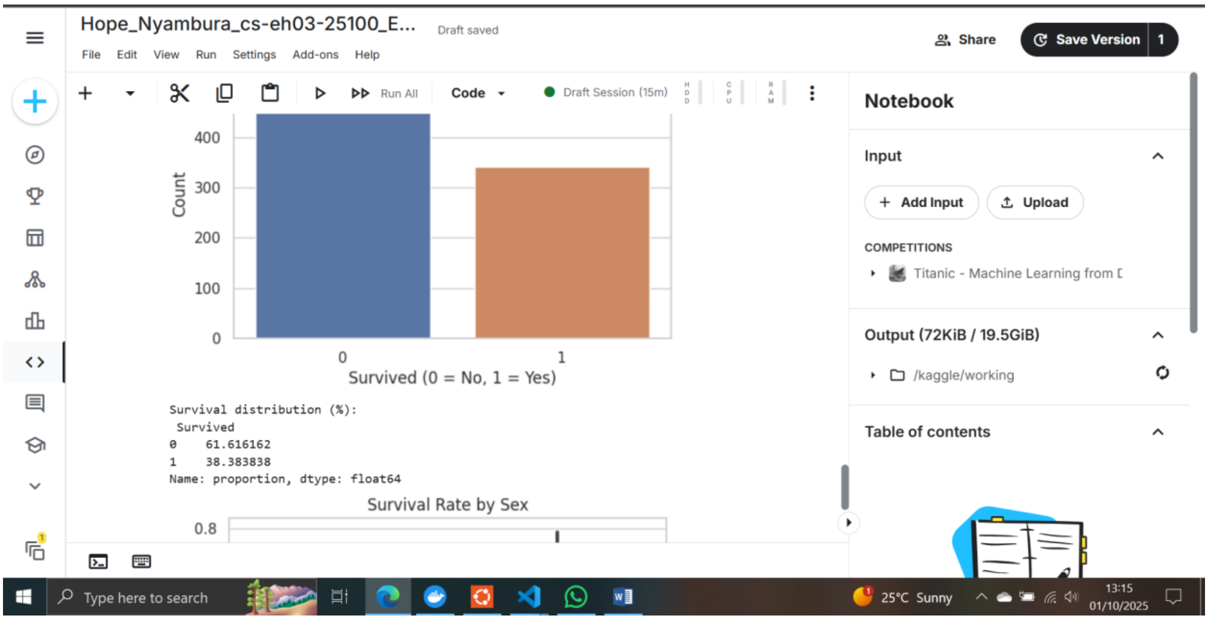


Step 7: Multivariate Analysis

- Pclass, Sex, and Survival → grouped bar plot.
- Pclass, Age, Fare, and Survival → scatter plots.
- Embarked, Pclass, and Survival → stacked plots.

Step 8: Target Variable Analysis

- Target = Survived (0 = No, 1 = Yes).
- Dataset is imbalanced (~62% did not survive, ~38% survived).
- Key findings:
  - Women had higher survival rates.
  - Higher class passengers had higher survival.
  - Younger children had higher chances of survival.



4. Conclusion

This assignment provided valuable practice in exploratory data analysis using Pandas, Matplotlib, and Seaborn.
Key insights include:

- Gender and class were strong predictors of survival.
- Fare and age showed important but less direct relationships.
- Embarkation point also influenced survival, particularly when combined with class. The dataset is now well-understood and ready for further predictive modeling.

5. Link to Kaggle Notebook

[https://www.kaggle.com/code/defhopeee/hope-nyambura-cs-eh03-25100-eda]